

Topology, the Complex Hopf Fibration, and Finite Approximations in HQIV:

TUFT Alignments, Contact Spectra, and the Discrete–Continuous Interface

Steven Ettinger Jr

Independent Researcher — steven@disregardfiat.tech

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Abstract

Position in the publication chain. This is a Tier-2 topology companion in the HQIV preprint series. It supplies the topological “why” layer for structures already present in the HQIV discrete null-lattice and Fano/octonion carrier: the rigidity of three generations, the shell-surface readouts, the effective continuous ξ chart, and the informational-energy mass row. The upstream anchors are the main HQIV framework [3], the light-cone null-lattice readout [5], and the $\mathfrak{so}(8)$ closure [7]. The electroweak/mass-spectrum paper [2] and the Standard-Model Lagrangian paper [1] are the intended sibling Tier-2 companions that will cross-cite this work once all records carry minted DOIs. The three published Tier-1 companions [8, 6, 4] provide the immediate dimensional, variational, and finite-mode spine.

Result. Jenny Lorraine Nielsen’s Topological Unified Field Theory (TUFT) on the complex Hopf fibration [9] proves that any $U(1)$ -complete gauge theory with discrete charge quantization must, up to homotopy and bundle isomorphism, be formulated on the universal Hopf fibration $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ and its finite approximations $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$. The Standard-Model gauge factors and a gravity sector with fiber-induced torsion arise as reductions along nested Hopf shells, with a contact Beltrami spectrum on each shell and zeta-regularized sector determinants supplying intrinsic mass scales. This paper records the precise, machine-checked alignments already present in HQIV_LEAN between TUFT’s finite-shell contact geometry and HQIV’s discrete null-lattice + Fano carrier, together with the explicit chart distinctions that keep the two frameworks from being silently identified. The 4/3 lock-in-neighbour match, the three-generation count from integrable fibre windings, the Beltrami versus Peter–Weyl normalisation distinction, and the placement of the 3/2 ratio in Fano holonomy rows rather than geometric resonance steps have all been formally verified in Lean without the use of ‘sorry’. These alignments are bidirectional: TUFT supplies a topological derivation for structures HQIV had taken as geometric axioms; HQIV’s patch ontology supplies the finite-truncation discipline that prevents either framework from treating $\mathbb{CP}^\infty$ or the continuous ξ chart as literal ontological commitments.

Patch-ontology reader contract. Per the patch-theory reader contract imported below, the discrete patch layer (shell indices $m \in \mathbb{N}$, finite index types $\text{Fin } n$, horizon-limited charts) is the ontology. Smooth-manifold notation, continuous ξ charts, and $\mathbb{CP}^n$ language appearing in this paper are translation layers for comparison with the TUFT literature and with standard physics. They are not claims that a fundamental continuum spacetime or an ultraviolet limit has been recovered or is required. Taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit on the load-bearing objects (lock-in at referenceM = 4, Fano holonomy rows, resonance steps, informational-energy localization) would erase the discrete index that does the predictive work.

Status. The alignments recorded here, including the typed [Hopf-shell complex + ContactBeltrami](#) layer (now part of the official ‘HQIVParallelPoincare’ and ‘HQIVPhysics’ build globs and verified green), the T10 three-row phase assembler with admissible-cycle and orientation fields, the T11 canonical matrix torsion with skew-adjoint action, the T12 concrete non-factorability witness, and especially the exact algebraic trapped-Casimir factorization of the binding cell together with the active per-shell curvature-imprint program, have been formally verified or constructed in Lean without the use of ‘sorry’. A dedicated section below articulates the principal contributions of Nielsen’s framework (the classifying-space necessity of the complex Hopf fibration for charge-quantized theories, the finite-approximation discipline, the contact Beltrami spectrum and fibre holonomy phases as a potential source for mixing phenomena, and the single-scale derivation programme, including a re-interpretation of apparent dark-sector effects as internal to HQIV’s dimensioned quantities at the present epoch). The paper further records the precise points at which the patch ontology of HQIV and the Lean formalization supply mutual reinforcement. Certain deeper alignments remain the subject of ongoing formalization efforts and are catalogued in the accompanying technical note on formalization targets. No universality theorem or single-Fermi-constant derivation is imported from Nielsen’s work. Reader-friendly anchors live in [Appendix A](#).

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. `Fin 4`), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (\mathbf{e}_1, \mathbf{e}_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from

discrete ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \mathrm{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1 Introduction

The HQIV framework is built on a discrete null-lattice whose combinatorial growth, Fano incidence structure, and horizon readouts already produce three-generation rigidity, shell-surface mass ladders, rational coupling ratios ($\alpha = 3/5$, $\gamma = 2/5$), and an effective continuous ξ chart without ever positing a fundamental smooth spacetime. At the same time, the topological literature contains a striking result, established by Nielsen: any gauge theory whose $U(1)$ sector is both complete—realising every principal $U(1)$ -bundle over an arbitrary paracompact base—and charge-quantised must be formulated, up to homotopy equivalence of the base and isomorphism of bundles, on the universal complex Hopf fibration $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ and its finite approximations $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$ [9].

Nielsen further demonstrates that the Standard-Model gauge factors arise as natural reductions along this nested shell hierarchy: the circular S^1 fibre yields the $U(1)$ electromagnetic sector, the first non-trivial shell S^3 furnishes $SU(2)$, and the subsequent shell S^5 accommodates $SU(3)$. Gravity emerges from the Kähler geometry of the base together with torsion induced by the fibre, producing a structure analogous to Einstein–Cartan theory in which the Levi-Civita connection is recovered precisely when the torsional degrees of freedom are set to zero. On each shell the generalised Beltrami operator $B = d|_\xi$, acting on the contact distribution, is shown to be elliptic and essentially self-adjoint, with a discrete spectrum that remains stable under torsion perturbations by the Kato–Rellich theorem [9].

Fibre-winding decomposition yields independent topological sectors whose Gaussian functional determinants, regularised via spectral zeta functions, generate intrinsic mass scales without additional dimensionful input beyond a single electroweak reference. Fermion mixing matrices are likewise derived from the intersection-form overlaps of admissible cycles in the cohomology of $\mathbb{CP}^4$, with CP-violating phases arising directly from fibre holonomy. Dynamics, finally, are traced to the fluctuation spectrum of the topological action on the outermost shell S^9 . These results furnish a unified geometric origin for both the particle spectrum and the gauge-gravity structure.

This paper presents a rigorous examination of the interface between Nielsen’s construction and the HQIV framework. It identifies the precise respects in which the nested Hopf shells, the contact Beltrami spectrum, the fibre-winding sectors, and the fibre-induced holonomy of the former illuminate structures that HQIV had previously derived from discrete combinatorial and variational principles. The examination is supported by machine-checked formalizations in the `HQIV.LEAN` library, which record both the alignments and the chart-specific distinctions necessary for phenomenological accuracy.

The central methodological point is the shared commitment to finite approximations. TUFT works with the finite Hopf shells $S^{2n+1} \rightarrow \mathbb{CP}^n$ and treats the infinite case as a direct limit; HQIV works with finite shells $m \in \mathbb{N}$ truncated at the lock-in horizon reference $M = 4$ and treats the continuous ξ chart as an effective readout generated by the discrete data. In both cases a literal continuum is not required and in several places would erase the very indices that carry the physics.

1.1 Limitations and Scope

The present study confines itself to those aspects of Nielsen’s framework that admit direct comparison with structures already present in the HQIV formalism—namely the finite Hopf-shell hierarchy, the contact Beltrami spectrum, fibre-winding multiplicity, and the geometric origin of effective mass and coupling scales. No claim is advanced that the full universality theorem, the single-Fermi-constant derivation of the entire spectrum, or the detailed construction of the topological action on S^9 have been reproduced or endorsed. The Lean development likewise remains partial: while the low-winding spectral alignments and the typed substrate for contact data are machine-verified, higher-order structures such as intersection-form mixing matrices and the explicit fluctuation spectrum on S^9 are still at the level of rigorously formulated open targets.

A useful complementary perspective is supplied by the analysis of three-dimensional causal growth on the HQIV null lattice [8]. That work isolates the quadratic shell-counting law $A(m) = 4(m+2)(m+1)$ as the combinatorial signature of 3D causal growth and packages a machine-checked conditional forcing theorem: once a shared-manifold rapidity bridge and an explicit $\mathfrak{so}(8)$ span witness are assumed as hypotheses, common rapidity, positivity and divergence of the curvature integral, and reference normalisation follow. The paper is explicit that a pure causal partial order does not, by itself, select the octonion multiplication table or the full internal gauge structure; the octonionic completion is conditional on additional geometric data. The same work develops a spanning-tree model of causal monogamy that derives the curvature-imprint pair $(\alpha, \gamma) = (3/5, 2/5)$ and identifies the Fano plane as the natural 3D projective spine for the seven imaginary directions. These results supply both positive combinatorial content for the target discrete 3-complex used in the typed Hopf-shell mapping and a clear methodological limit: even within the HQIV framework, certain topological features require further structure beyond pure causal growth. This conditional character provides a natural counterpoint to the stronger necessity claims advanced in Nielsen’s construction and helps delineate the precise scope of any combined programme.

2 The finite-approximation discipline

TUFT’s construction begins with the observation that the classifying space for $U(1)$ -bundles with discrete charge is the infinite complex projective space $\mathbb{CP}^\infty$, with the universal bundle the infinite Hopf fibration $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$. All finite approximations $S^1 \rightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$ are admissible truncations. The Standard-Model factors arise as successive reductions: $U(1)$ from the circular fiber, $SU(2)$ from the S^3 shell, $SU(3)$ from the S^5 shell. Gravity emerges from the Kähler geometry of the base together with torsion induced by the fiber.

HQIV never begins from a bundle classification. It begins from a discrete null-lattice whose vertices are tagged by stars-and-bars at each shell m , whose incidence is governed by the Fano plane, and whose horizon areas $S(m) = (m+1)(m+2)$ supply the surfaces that appear in every mass and coupling readout. Three generations appear because the integrable fiber-winding sectors that survive in the discrete complex are exactly three; the lock-in at $m = 4$ is a combinatorial closure condition, not a fitted vev.

The two pictures meet at the level of finite truncations. The TUFT statement that only $n = 1, 2, 3$ yield integrable torus sectors (unknot, Hopf link, trefoil) before a hyperbolic transition at $n = 4$ supplies a topological reason for the HQIV fact that ResonanceGeneration is exactly Fin3. The finite Hopf shells supply a reason why a discrete shell ladder with a distinguished lock-in horizon should exist at all. Conversely, the HQIV patch ontology supplies the rule that neither $\mathbb{CP}^\infty$ nor the continuous ξ chart may be treated as fundamental; they are limiting readouts of finite data.

3 Contributions of Nielsen’s Topological Framework to HQIV

The value of the TUFT perspective is not that it supplies new numerical fits, but that it supplies a topological *necessity* argument and a spectral-geometric origin story for structures that HQIV had previously derived from discrete combinatorial and variational axioms. The most substantial contributions fall under four headings.

3.1 Classifying-space necessity and indecomposability

TUFT proves that any gauge theory whose $U(1)$ sector is both complete (realises every principal $U(1)$ -bundle over paracompact bases) and charge-quantised (admits only discrete charges) must, up to homotopy equivalence and bundle isomorphism, be formulated on the universal complex Hopf fibration and its finite approximations. The resulting total structure group $G_{\text{total}} = (SU(3) \times SU(2) \times U(1) \times SO(4))/\Gamma$ is intrinsically non-factorable because the first Chern class c_1 generates the cohomology ring $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \simeq \mathbb{Z}[c_1]$.

In the HQIV setting this supplies a principled reason why the Fano plane, the octonion carrier, and the three-generation count appear together rather than as independent modelling choices. The discrete null-lattice with its seven-line incidence structure can now be read as a combinatorial model of the admissible reductions along the first few Hopf shells. The non-factorability claim gives additional weight to the already-proved algebraic closure results that generate $\mathfrak{so}(8)$ from $G_2 \cup \{\Delta\}$ inside the same eight-dimensional carrier.

3.2 Finite approximations and the “no infinities” discipline

Both frameworks are committed to finite truncations. TUFT works with $S^{2n+1} \rightarrow \mathbb{CP}^n$ and treats $\mathbb{CP}^\infty$ as a direct limit; HQIV works with shells $m \in \mathbb{N}$ truncated at the lock-in horizon reference $M = 4$ and treats the continuous ξ chart as an effective readout. The TUFT statement that only windings $n = 1, 2, 3$ yield integrable torus sectors before a hyperbolic transition at $n = 4$ furnishes an independent topological reason for the HQIV fact that ResonanceGeneration is exactly Fin3. Conversely, the HQIV insistence that continuum limits are information-destroying readouts (rather than ontological improvements) supplies the methodological safeguard that prevents the TUFT infinite fibration from being misread as a commitment to a fundamental smooth spacetime.

3.3 Contact spectrum, holonomy phases, and mixing

TUFT’s generalised Beltrami operator on the contact distribution of each Hopf shell yields a discrete spectrum stable under torsion perturbations (Kato–Rellich). The Lean formalisation in [HopfShellComplex](#) together with the earlier bridges records both the shared eigenvalue law $\lambda_\ell = \ell(\ell + 2)$ on S^3 and the explicit normalisation distinction between TUFT’s fundamental coexact mode and the scalar Peter–Weyl value. The fibre-winding decomposition supplies independent topological sectors whose zeta-regularised Gaussian determinants are intended to generate intrinsic mass scales; the existing HQIV informational-energy and sector-Gaussian-leading-weight constructions are now seen as discrete, patch-level realisations of precisely this mechanism.

Fermion mixing (CKM/PMNS) is claimed to arise from intersection-form overlaps of admissible cycles in $H^*(\mathbb{CP}^4)$, with CP violation induced by fibre holonomy phases. The rh-fourier-lift curvature channel $K(n, \alpha)$ and its associated PhaseMap (already wired to the new Hopf-shell holonomy carrier) provide a concrete discrete realisation of those phases. The existing Fano holonomy rows and imprint phase machinery therefore acquire a possible topological interpretation rather than remaining purely phenomenological charts.

3.4 Single-scale derivation versus multi-witness architecture

TUFT asserts that a single empirical input—the Fermi constant together with its associated electroweak vacuum expectation value—suffices to determine the fine-structure constant and all shell-specific mass scales, spectral coefficients, and couplings via the spectral geometry of the fibration. HQIV, by contrast, maintains an explicitly multi-witness architecture (proton lock-in, CODATA α , present-CMB) with operational discipline about which scale is active in any given solve. Nielsen’s framework therefore presents a research programme that invites comparison with HQIV’s current multi-witness architecture. The machine-checked chart distinctions recorded in the Lean library (the $4/3$ alignment at the lock-in neighbour under the $m = n + 1$ chart, the location of the $3/2$ ratio within Fano holonomy rows rather than geometric resonance steps, and the separation of lepton resonance ratios from the low-winding Beltrami ratios) provide the necessary precision for any such comparison.

The new typed layer `Hqiv.Topology.HopfShellComplex` (with its functor to the discrete 3-complex substrate and its `PhaseMap` carrier hook) makes several of the above alignments machine-checkable rather than narrative. It does not yet reproduce the full contact geometry, Ray–Singer torsion, or the universality theorem; those remain explicitly open targets.

3.5 Re-interpretation of TUFT’s dark sector

TUFT lists “dark sector effects from bundle torsion and holonomy” among the consequences of its construction. HQIV has no separate dark sector by construction: all energy-density contributions are internal to the informational vacuum of the discrete null-lattice plus its curvature and phase structure.

The precise correspondence is as follows. TUFT derives mass scales and apparent extra contributions from the zeta-regularised Gaussian determinants of its fibre-winding sectors together with torsion and holonomy corrections on the Hopf fibration. In HQIV the dimensioned quantities observed at the present epoch (“now”) are given by the informational energy

$$E_{\text{tot}}(m_{\text{rest}}, \xi_{\text{now}}) = m_{\text{rest}} + \frac{1}{\Theta_{\text{local}}(\xi_{\text{now}})}, \quad \Theta_{\text{local}}(\xi) = \frac{T_{\text{Pl}}}{\xi},$$

together with additive corrections from the curvature channel $K(n, \alpha)$ (whose $\alpha \log x$ term supplies the imprintable geometric contribution) and the rapidity/phase lifts already wired to the Hopf-shell holonomy carrier.

Thus TUFT’s “derived constants” (mass scales, effective couplings, spectral coefficients obtained from a single Fermi-scale input via Hopf spectral geometry) are exactly the effective values of HQIV’s dimensioned now quantities evaluated at the current horizon coordinate ξ_{now} (or the corresponding lock-in or outer-horizon shell in the discrete ladder), including the informational localization and curvature-imprint terms. What appears in TUFT as a dark sector is, under the HQIV reading, the internal energetic contribution of the informational vacuum itself once it is dimensionally evaluated at now. No additional fields or sectors are required.

A concrete experimental discriminant arises from the contrasting origins assigned to apparent new effects. Nielsen’s framework predicts a torsion-induced phase wobble generated by fibre holonomy, detectable in precision interferometry or spin-precession measurements. The HQIV account instead attributes analogous signatures to the curvature-channel imprint: the $\alpha \log x$ term in the discrete density $\rho(x)$ accumulates into $K(n, \alpha)$ and modulates effective couplings at the present epoch. Distinguishing global fibre-holonomy wobble from local curvature-imprint modulation therefore provides a sharp test accessible to existing and near-future precision experiments. A quantitative

estimate of the expected wobble amplitude (for example in current-generation atom interferometers or spin-precession measurements) would render the discriminant still more incisive; such an estimate is not yet available within the present formalisation and is reserved for subsequent work.

This re-interpretation sharpens the single-scale versus multi-witness comparison: the additional contributions that must be derived geometrically in Nielsen’s framework appear in HQIV as the explicit $1/\Theta$ localisation and curvature-channel terms already present in the definition of observable quantities at the present epoch. The Lean formalisation of the informational-energy readouts and the curvature channel $K(n, \alpha)$ therefore provides a direct computational bridge for quantitative comparison.

4 Proved Lean alignments (the honest bridge)

All alignments below are stated as theorems that have been formally verified in Lean without the use of ‘sorry’. They are deliberately chart-specific; the library contains explicit mismatch theorems wherever a naïve identification would be false.

4.1 Beltrami spectrum and normalization distinction

The scalar Laplace–Beltrami eigenvalues on the unit S^3 are $\lambda_\ell = \ell(\ell + 2)$. TUFT’s coexact Beltrami operator on the contact distribution yields the same law. The Lean library records both normalizations and proves they differ at the fundamental level:

- `beltramiPeterWeylEigenvalueS3 ℓ = laplaceBeltramiEigenvalueS3 ℓ` (shared law).
- The TUFT fundamental coexact mode uses $\lambda_1 = 2$; the Peter–Weyl level $\ell = 1$ gives 3. The theorem `tuftFundamentalBeltrami_ne_eq_peterWeyl_one` records the mismatch explicitly.

Multiplicity on the S^3 shell is $(n + 1)^2$ for winding n , matching TUFT sector multiplicity.

4.2 Three generations from integrable fiber windings

TUFT states that only the first three positive fiber windings yield integrable torus sectors. The Lean definition

`HopfFiberWinding n : Prop`

is true precisely for $n = 1, 2, 3$ and false for $n = 0$ and $n \geq 4$. The theorems `hopfIntegrableGenerationCount_eq_three` and `tuftMinimalBeltrami_matches_fin3` establish that this counting is identical to the HQIV `ResonanceGeneration = Fin 3` and that the Beltrami eigenvalues are strictly ordered on those three slots.

4.3 Lock-in neighbour match and chart distinctions

Under the bookkeeping chart $m = n + 1$ (TUFT winding to HQIV discrete shell), the minimal Beltrami ratio for windings $3 \rightarrow 2$ is exactly $4/3$. On the HQIV side the geometric resonance step at the lock-in neighbour shells is

$$\text{geometricResonanceStep } 4\,3 = 4/3.$$

The equality `tuftBeltrami_tau_mu_eq_geometricResonance_lockin_neighbor` is a proved theorem.

The same machinery records the mismatch at the previous step: the Beltrami ratio $2 \rightarrow 1$ is $3/2$, but the geometric step on shells $3 \rightarrow 2$ is $35/24$. The $3/2$ ratio does appear, but in the Fano holonomy rows (`holonomyRowRhs` on the heavy and middle vertices) rather than in the geometric resonance steps. These distinctions are not bugs; they are the content of the honest bridge.

4.4 Informational energy and spectral weight scaffold

TUFT obtains intrinsic scales from zeta-regularized Gaussian determinants over the fiber-winding sectors. HQIV already possesses an informational energy

$$E_{\text{tot}}(m_{\text{rest}}, \xi) = m_{\text{rest}} + \frac{1}{\Theta(\xi)}$$

and a Beltrami-level correction term `beltramiSpectralWeightS3` $\ell = (\text{beltramiPeterWeylEigenvalueS3} \ell + 1)^{-1}$. The scaffold `informationalEnergyAtXiWithBeltrami` packages the sum. The deeper identification of this term with a per-sector zeta determinant remains open (target T8).

5 Discrete topology pegs already present

The library contains a discrete 3-complex layer (`Hqiv/Topology/DiscreteNullLatticeComplex.lean`) with `NullShellVertex`, `Discrete3Complex`, Euler characteristic, combinatorial sphericity ($\chi = 0$ for S^3 -like references), and an `S3NullReference` template whose edges and triangles are currently empty. This is the natural host for Hopf-shell contact data or fiber-winding sectors. The new [Hopf-shell complex module](#) supplies the typed ‘HopfShell’ + ‘ContactBeltrami’ layer that maps directly onto this substrate (T6/T7).

Parallel modules (`ParallelPoincareScaffold`, `DiscretePhaseEvolution`, `SignedShellBudget`) already carry discrete Poincaré and phase-evolution language compatible with torsion and holonomy without continuum commitments.

6 Synergy with the rh-fourier-lift curvature channel

On the current development branch the curvature density $\rho(x, \alpha) = (1 + \alpha \log x)/x$ and its cumulative channel $K(n, \alpha)$ are proved to dominate the harmonic series, to be strictly monotone for $\alpha > 0$, and to admit a phase lift `PhaseMap` (canonical example: rigid rotation by $\log \omega$). This discrete phase-lift mechanism is the natural carrier for TUFT fiber holonomy phases. The same channel supplies a concrete, finite-n model for the torsion that TUFT derives from the fiber in the Einstein–Cartan-like sector.

7 Exact trapped-Casimir factorization of the binding cell

A major recent formal advance is the explicit algebraic factorization of the existing HQIV binding law as a lattice-counted trapped zero-point (Casimir) energy term plus one normalized selection factor drawn from the $\text{SO}(8)$ carrier:

$$E_{\text{bind}} = \text{lattice count} \times (\text{per-mode zero-point}) \times \text{normalized SO}(8) \text{ trace selection.}$$

In Lean this is recorded by the definitions ‘`normalizedSO8TraceSelection`’ and ‘`trappedCasimirCouplingCell`’, with proved identities ‘`trappedCasimirCouplingCell_eq_alphaEffAtShell`’ and ‘`bindingCouplingAtShell`’.

The T11/T12 layer now supplies a named closure predicate ‘T11T12ClosesTrappedCasimirSelection’ (a Hopf shell’s torsion coefficient supplies exactly the required normalized selection at a given readout shell) together with a conditional theorem that turns the ontology into a precise proof obligation. The T12 three-shell non-factorability witness is wired directly into this coefficient.

This work does not yet claim that “gluons do not exist.” It isolates the residual as a sharply defined selection factor on the closed carrier and makes the statement “the T11/T12 torsion data on the integrable Hopf shells reproduces that selection” a theorem target rather than a narrative hope. The per-shell ‘curvatureImprintAlpha’ hook (with the associated ‘trappingSelectionFromHeavyHopf-ShellWithAlpha’ and three-shell variants) supplies the natural geometric dial for exploring whether different contact geometries on the successive Hopf shells can furnish different effective imprints that move both stabilization horizons and trapping factors.

8 Open Phase-2 targets

Substantial machine-checked progress has been made on the topology-specific targets T6–T12 (typed Hopf-shell complexes with direct mapping into the null-lattice 3-complex, ContactBeltrami spectrum and formal stability scaffolding, per-sector zeta-determinant leading-term witnesses, T10 phase-contribution assembler with full three-row coverage including the light slot together with admissible-cycle and orientation fields, T11 canonical matrix torsion with skew-adjoint action and Parallel-Poincaré bridge, and T12 concrete non-factorability witness with closure condition). The exact trapped-Casimir factorization of the binding cell (see above) and the active per-shell curvature-imprint program (‘curvatureImprintAlpha’ hook + explicit ‘trappingSelection*WithAlpha’ objects) constitute the current research frontier.

The mass-spectrum targets T1–T5 remain open, though T1 now possesses strong T10/T12 interlocks and the MeV chart layer has been made fully parametric. The deeper items on T13 (*S⁹ fluctuations*) and several Phase – 3 topics (*real admissible – cycle overlap forms, genuine CP – phase threading, full contact operator on forms*) are still at the scaffold early-theorem stage. All of these targets are in the input derivation of the full spectrum.

9 Conclusion

TUFT supplies a topological derivation for the existence and structure of the discrete shells, the three-generation count, the contact spectrum, and the fiber holonomy that appear in HQIV as geometric or combinatorial facts. HQIV supplies the patch-ontology discipline and the finite-truncation rule that keep both pictures from treating their continuous limiting objects as fundamental. The Lean library already certifies the precise points of alignment and the precise points of chart mismatch. The deeper program—populating the discrete complex with Hopf data, lifting the contact Beltrami operator, and exhibiting the curvature channel as a discrete torsion source—is now a well-scoped, bidirectional research program rather than a narrative hope.

A Lean module index

The following reader-friendly names appear as hyperlinked anchors in the body. Each entry records the Lean module and the principal identifier(s).

Beltrami–Peter–Weyl eigenvalue law. Shared spectral law on unit S^3 . Lean: `beltramiPeterWeylEigenvalue`
`beltramiPeterWeylEigenvalueS3_eq_laplace` in [Hqiv/Physics/HopfShellBeltramiMassBridge](#).

[lean](#).

TUFT fundamental vs. Peter–Weyl normalization. Explicit mismatch at the fundamental level. Lean: `tuftFundamentalBeltrami_ne_eq_peterWeyl_one` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Hopf fiber winding (integrable sectors). Exactly three positive windings yield integrable torus sectors. Lean: `HopfFiberWinding`, `hopfIntegrableGenerationCount_eq_three`, `tuftMinimalBeltrami_matches_fin3` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Minimal Beltrami eigenvalue per winding. $\lambda_{\min}(n) = n+1$. Lean: `tuftMinimalBeltramiEigenvalue` and the one/two/three evaluation theorems in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

TUFT Beltrami resonance ratios. $3 \rightarrow 2$ gives $4/3$; $2 \rightarrow 1$ gives $3/2$. Lean: `tuftBeltramiResonanceRatio_tau_mu`, `tuftBeltramiResonanceRatio_mu_e` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Lock-in neighbour $4/3$ alignment. Under the $m = n + 1$ chart the Beltrami ratio equals the geometric resonance step at the lock-in neighbour shells. Lean: `tuftBeltrami_tau_mu_eq_geometricResonance_lockin_neighbor`, `geometricResonanceStep_four_three_eq_four_thirds` in [Hqiv/Physics/HalfStepBeltramiShellBridge.lean](#).

Explicit chart mismatch at $2 \rightarrow 1$. Beltrami $3/2$ is not the geometric step $35/24$ on shells $3 \rightarrow 2$. Lean: `tuftBeltrami_mu_e_ne_geometricResonanceStep_three_two` in [Hqiv/Physics/HalfStepBeltramiShellBridge.lean](#).

$3/2$ ratio in Fano holonomy rows. The Beltrami $2 \rightarrow 1$ ratio equals the holonomy-row budget ratio on the heavy and middle Fano vertices. Lean: `tuftBeltrami_mu_e_eq_holonomyRowRhs_vertices`, `tuftBeltrami_mu_e_eq_fanoHolonomyWeight_ratio` in [Hqiv/Physics/HalfStepBeltramiShellBridge.lean](#).

Sector Gaussian leading weight. `detunedShellSurface = shellSurface/omaxwellFanoDetuning1Jet`. Lean: `sectorGaussianLeadingWeight_eq_detunedShellSurface` in [Hqiv/Physics/FanoSectorSpectralMassEmergence.lean](#).

Laplace–Beltrami on S^3 and S^4 . Eigenvalue laws and dimension formulas. Lean: [Hqiv/Geometry/QuaternionMaxwellS30MaxwellS4Spectral.lean](#).

Discrete 3-complex layer. `NullShellVertex`, `Discrete3Complex`, Euler characteristic, combinatorial sphericity, `S3NullReference` template. Lean: [Hqiv/Topology/DiscreteNullLatticeComplex.lean](#).

Curvature channel and phase map (rh-fourier-lift). $\rho(x, \alpha)$, $K(n, \alpha)$ with proved domination and monotonicity; `PhaseMap` abstraction with canonical $\log\omega$ lift. Lean: [RhFourierLift/Setup.lean](#).

Hopf-shell complex + ContactBeltrami (T6/T7 core). Typed ‘`HopfShell`’ with integrable-winding certificate, functor ‘`toDiscrete3Complex_integrable_landing_in_the_null – lattice3 – complex_substrate`’, ‘`ContactBeltrami`’ spectrum and multiplicity agreement with existing Beltrami data, and ‘`Holonomy`’.

fourier – lift‘PhaseMap‘.Lean : . Now part of the official ‘HQIVParallelPoincare‘ and ‘HQIV-Physics‘ build globs; imported by ‘ShellOpeningEvolution‘ (T9 wiring) and the Beltrami bridge (compatibility transport).

Scaffold adding the inverse Beltrami weight to the localization term. Lean: `informationalEnergyAtXiWithBeltramiBeltramiSpectralWeightS3` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Explicit objects for exploring different effective curvature imprints α_n on the integrable Hopf shells. Lean: `trappingSelectionFromHeavyHopfShell`, `trappingSelectionFromHeavyHopfShellWithAlpha`, `trappingSelectionFromThreeHopfShellsWithAlphas` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

Algebraic factorization of the binding cell as lattice-counted zero-point energy plus normalized SO(8) selection. Lean: `normalizedS08TraceSelection`, `trappedCasimirCouplingCell`, `trappedCasimirCouplingCell_eq_alphaEffAtShell`, `bindingCouplingAtShell_eq_lattice_trappedCasimirCell`, `T11T12ClosesTrappedCasimirSelection`, `bindingCouplingAtShell_eq_T11T12_trappedCasimir` in [Hqiv/Physics/HopfShellBeltramiMassBridge.lean](#).

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